

ME 127: Final Portfolio

Michael Beals
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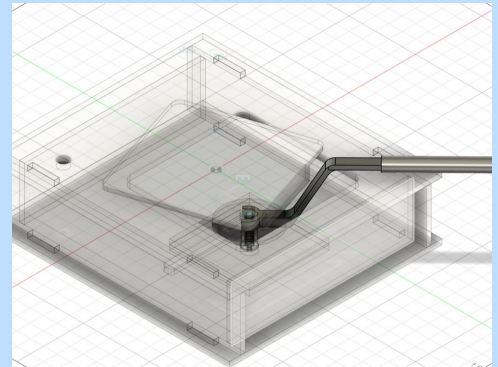
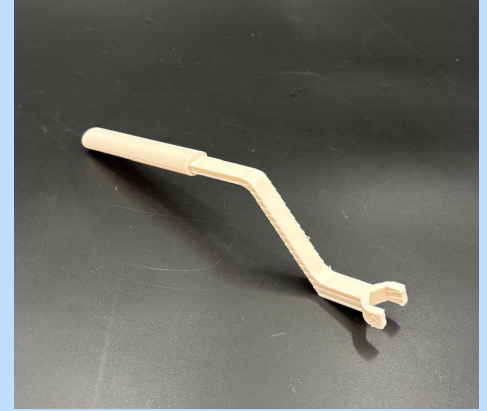
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Algorithmic Design

Key Takeaways

Project 1: Space Wrench

In our first project, we were asked to design a wrench that could be used in space and manufactured additively. The challenge was deceptively simple: how do you ensure function, strength, and printability in a weight-sensitive environment? I learned that details like chamfers, overhangs, and support structure design were critical to print success. I went through several iterations, adjusting geometry to ensure my wrench could withstand a 30 lbf load in simulation while minimizing material use. From this project, I learned how closely tied AM design is to print orientation and geometry decisions.



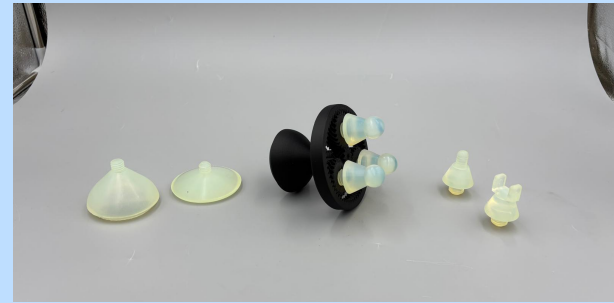
Project 2: AM Assemblies

For this project, I set out to design a remote controller that was both functional and physically satisfying to hold. I experimented with multiple AM assembly techniques, including a print-in-place joystick, a spinning volume dial, and a magnet-based button mechanism. I also integrated a mid-print insert to weight the controller, although I later pivoted to prioritize functional features. A key takeaway was the importance of prototyping and iteration—my initial ideas were too bulky and lacked purpose, but test prints helped refine tolerances and ergonomics. In the end, I created a remote that combined playfulness, tactile feedback, and mechanical nuance, and I gained confidence in tools like support blockers and the Bambu slicer to enhance print outcomes.



Project 3: Applied Materials

MyoFit was a collaborative project where our team designed a hand-held massager that functioned entirely mechanically and leveraged the complementary properties of two additive materials. I helped design and print the handle using Markforged's Onyx, focusing on toughness and precision for the gear components. Our planetary gear system allowed for consistent rotation under applied pressure, while the soft Durable Resin nubs ensured comfortable skin contact. I learned how critical material selection is to real-world performance—PLA cracked under stress, while Onyx flexed without failure. Despite being sick for part of the project, I took charge of final printing and assembly, and the process taught me how to optimize designs for support-free printing and reliable modular assembly across printers and materials.



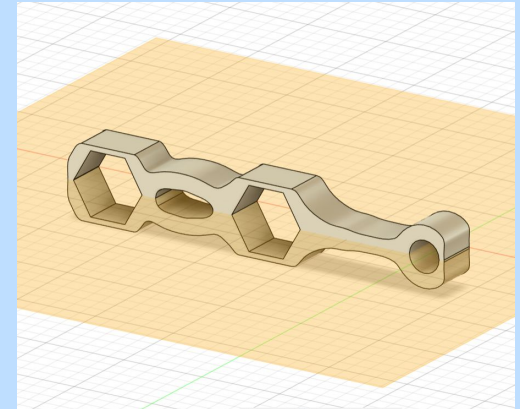
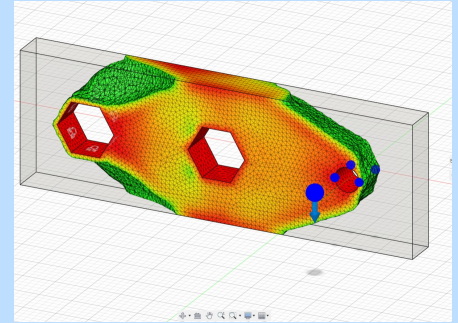
Project 4: AM Applications

This was by far my most personal and playful project. I created a multi-purpose organizer to hold my keys, sunglasses, AirPods, and water bottle—items I frequently misplaced. The design used generative design in Fusion 360 to resist tipping under asymmetric loads, especially with a full water bottle placed at an angle. The standout feature was the use of TPU noses, modeled from a scan of my own face, to hold sunglasses securely without slippage. PLA provided the rigid base structure, while TPU's higher coefficient of friction offered better grip. This project reminded me that engineering can intersect with humor, identity, and daily habit.



Algorithmic Design — Vise Speed Handle (HW4)

In this assignment, I redesigned a vise handle using both shape optimization and generative design. I ran simulations with realistic loads, modeling pushing, pulling, and even accidental impacts. My shape optimization reduced mass while maintaining strength, but generative design offered more creative, lightweight solutions. I discovered the importance of aligning manufacturing constraints with material selection—some combinations in Fusion 360 led to impossible results, like stainless steel additive designs. Ultimately, this assignment helped me understand when to use each tool: shape optimization for fast iteration, and generative design for broader exploration.



Key Takeaways

1. Iteration is a tool, not a sign of failure. Nearly every project I completed involved failed prints or reworked geometries. These were moments of growth, not mistakes.
2. Material choice matters more than I expected. PLA vs. TPU vs. Onyx each forced me to rethink design from both the user's and printer's perspective.
3. Design for AM is about constraint-based creativity. Support blockers, slicer tricks, and part consolidation all pushed me to rethink my default design instincts.
4. Generative tools are powerful, but only if paired with real constraints. Without understanding loads, material limits, or print capabilities, the results can be unprintable or impractical.

Final Reflections

ME127 was a course that constantly challenged me to merge form and function. Through each project, I came to see AM not just as a tool for rapid prototyping, but as a design language with its own grammar and constraints. I learned to embrace failure as feedback, to model with manufacturing in mind, and to find joy in making things that are both useful and a little ridiculous. Design is a long process, and prototyping is vital to success. Whether designing a vise handle or printing my own nose, I left this course with a deeper understanding of how engineering, design, and personality can coexist in every layer of a print.

